A thorough review of existing literature and systems was conducted to understand the current landscape of agricultural waste management and related technologies.

3.1 Agricultural Waste Management in India

Jain et al. (2014) estimated that approximately 92 million tonnes of crop residue are burned annually in India, contributing significantly to air pollution. Their study highlighted the need for technology-driven alternatives to open burning. The Indian Agricultural Research Institute (IARI) has documented the environmental and health impacts of stubble burning across multiple studies.

3.2 Digital Platforms for Waste Management

Several digital platforms have emerged for general waste management. Platforms like "Swachhata" by the Ministry of Housing focus on urban waste, while "mKisan" provides agricultural advisory services. However, none specifically address the agricultural waste tracking and recycling workflow that AgriTrace targets. The gap between general-purpose waste management apps and agricultural-specific requirements motivated the development of AgriTrace.

3.3 Firebase and Real-Time Database Systems

Google Firebase provides a comprehensive suite of backend services including Firestore (NoSQL database), Authentication, and Cloud Storage. Studies by Moroney (2017) have demonstrated Firebase's effectiveness in building real-time applications with offline support. The security rules system provides granular access control, which is critical for role-based applications like AgriTrace.

3.4 Next.js and Modern Web Frameworks

Next.js, developed by Vercel, has become the leading React framework for production applications. Version 15.5 introduces Turbopack for faster development builds and the App Router for improved routing and server components. TypeScript integration ensures type safety across the entire codebase.

3.5 Carbon Credit Systems

The carbon credit market has evolved significantly with the Paris Agreement. Studies by Pandey and Agrawal (2014) have quantified the carbon emissions from crop residue burning in India. AgriTrace uses scientifically established emission factors to calculate carbon offsets, with values ranging from 980 to 1400 kg CO_2 per metric tonne depending on waste type.